Data-Driven Decision Support for Evaluating AI-Generated Advertising Content in a Simulated Content Platform

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Academic research poster for an ESE 527 advertising analytics project. The work combines KPI monitoring, anomaly diagnosis, simulated A/B evaluation, interpretable prediction, and nested cross-validation into one decision-support workflow.

97,200 OBSERVATIONS | **20** VARIABLES | **0** MISSING / DUPLICATE ROWS

Background / Motivation

AI-generated advertising content can scale creative production, but small businesses still need a structured way to evaluate whether content performs well across contexts.

The project frames analytics as decision support: monitor platform health, detect abnormal movement, diagnose likely drivers, compare experiment variants, and estimate which contexts are more likely to produce high conversion performance.

Objective / Research Question

- Can a simulated content-platform workflow support decisions about which content contexts are more likely to perform well?
- Can descriptive analytics and predictive modeling be combined into one practical decision-support workflow?

Data Description

Observations	97,200	Variables	20.0000
Numeric variables	12.0000	Categorical variables	8.0000
Time variables	1.0000	Missing values	0.0000
Duplicate rows	0.0000	Data type	Simulated platform data

Rows combine content category, traffic source, user type, city tier, experiment group, KPI fields, and time-related context.

Methods / Workflow

1. KPI monitoring
Track funnel health and rate measures over time.

2. Anomaly detection
Flag unusual KPI changes for follow-up diagnosis.

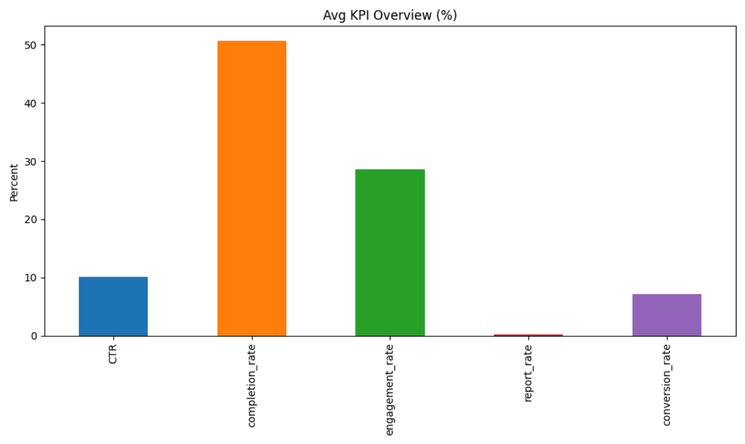
3. Segmented diagnosis
Compare content, traffic, user, and time segments.

4. Simulated A/B evaluation
Review experiment groups on business metrics.

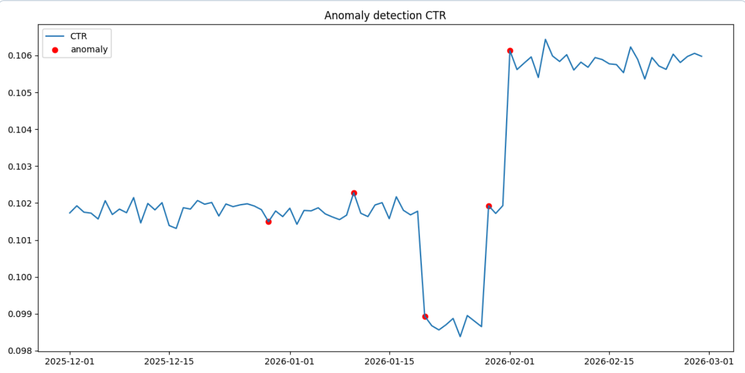
5. Predictive modeling
Classify high conversion performance with leakage controls.

6. Nested CV
Tune models and select features inside validation folds.

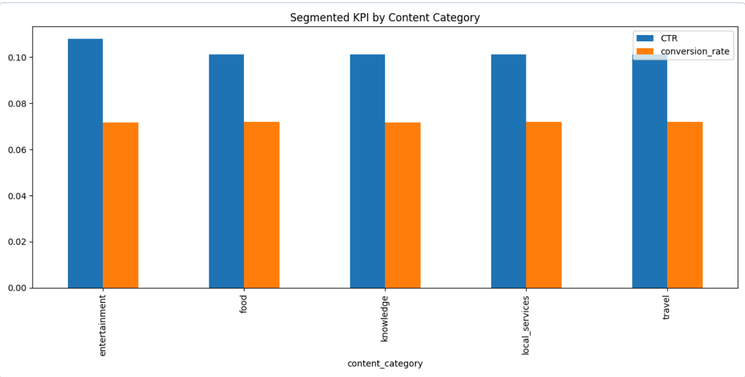
Descriptive Analytics Results



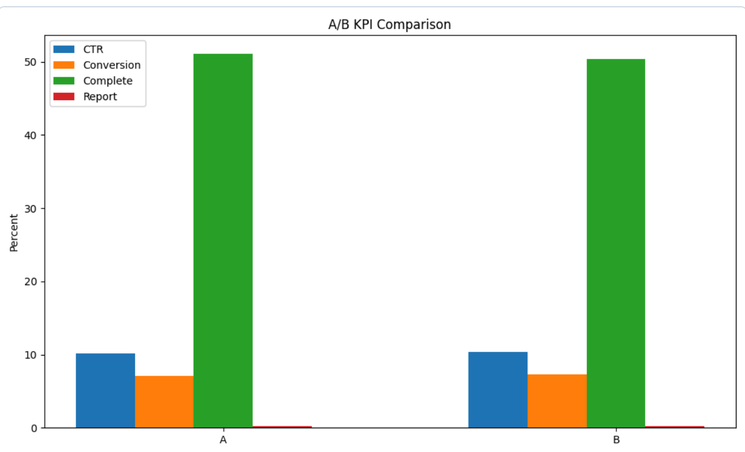
Dashboard overview



Anomaly alerts



Segmented diagnosis



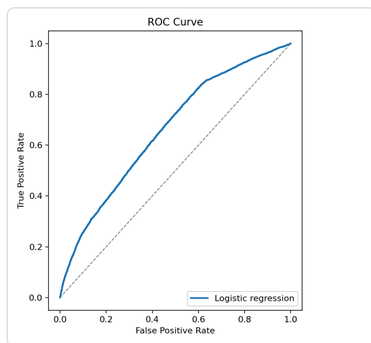
Simulated A/B comparison

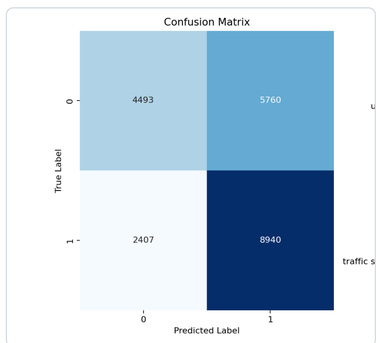
Experiment	CTR	Conversion Rate	Conversions
Group A	0.1015	0.0710	579,283
Group B	0.1040	0.0726	606,707

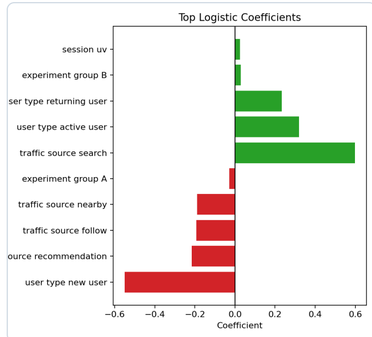
The descriptive layer supports monitoring, early warning, diagnosis, and simulated experiment review before predictive modeling.

Predictive Modeling Results

The predictive task is binary classification of high conversion performance. The target is defined using the training-set median conversion rate; feature inputs cover content, traffic, user, experiment, time, and session context. Downstream leakage variables are excluded.







ROC summarizes discrimination across thresholds. The confusion matrix shows classification outcomes at one threshold. Coefficients translate the model into directional business signals.

Logistic Regression result panels: ROC curve, confusion matrix, and top coefficients.

0.6219	0.6082	0.7879	0.6865	0.6606
Accuracy	Precision	Recall	F1-score	ROC-AUC

Coefficient Interpretation

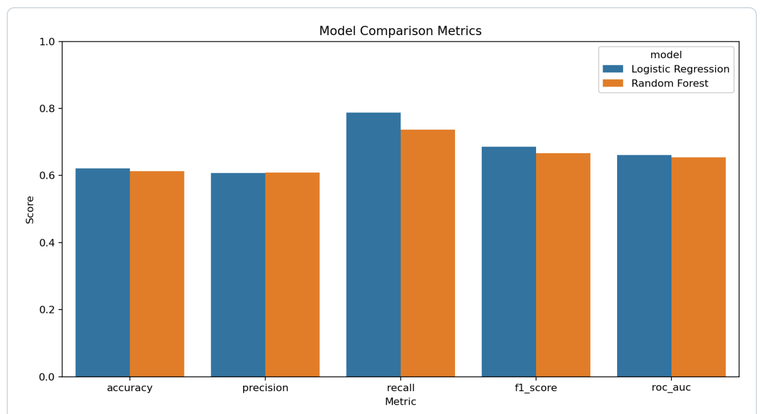
Higher predicted conversion association
- Search traffic
- Active users
- Returning users
- Experiment group B
- Session UV

Lower predicted conversion association
- New users
- Recommendation traffic
- Follow traffic
- Nearby traffic
- Experiment group A

Coefficients are directional associations in simulated data. They support business hypotheses, not causal claims.

Initial Model Comparison

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.6219	0.6082	0.7879	0.6865	0.6606
Random Forest	0.6133	0.6090	0.7371	0.6670	0.6536

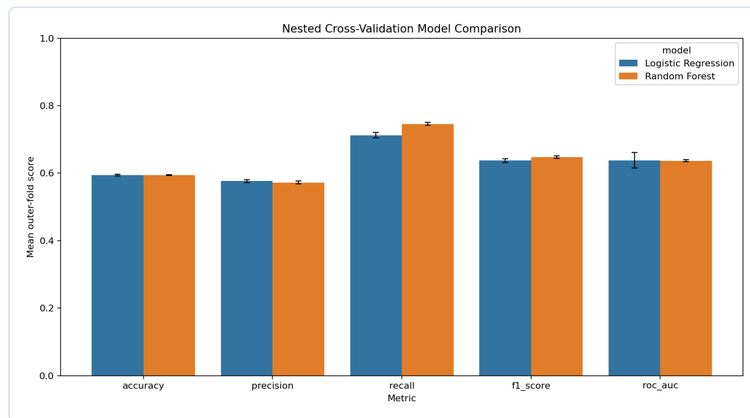


Initial split comparison: Logistic Regression is slightly stronger on ROC-AUC and remains easier to explain; Random Forest is a useful benchmark.

Validated Modeling Setup

Outer CV	3 shuffled KFold folds
Inner CV	3 StratifiedKFold folds
Tuning metric	ROC-AUC
Feature selection	SelectKBest inside the pipeline
Target threshold	Computed only from each outer training fold

Model Comparison and Validation



Nested CV is the strongest validation figure because model tuning and feature selection occur inside the validation workflow.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.5940	0.5767	0.7126	0.6372	0.6383
Random Forest	0.5939	0.5724	0.7458	0.6477	0.6367

Validation interpretation: Logistic Regression has slightly better ROC-AUC. Random Forest has slightly better recall and F1-score. The difference is small; the main improvement is stronger validation logic, not dramatic metric gains.

Key Findings

1. The project integrates descriptive, diagnostic, experimental, and predictive analytics into one workflow.
2. Search traffic and established user types are more associated with high conversion performance.
3. Logistic Regression remains the most interpretable predictive baseline.
4. Random Forest is useful as a benchmark but is not clearly better.
5. Nested CV improves rigor and reduces the risk of over-optimistic evaluation.

Limitations

- Source data is simulated, not live operational data.
- The A/B module is simulated rather than a real deployed experiment.
- Predictive models are prototypes and require external validation.
- Coefficients show association, not causation.

Conclusion / Next Steps

The project moved from monitoring and diagnosis toward an initial predictive workflow for evaluating AI-generated advertising content. Nested cross-validation strengthened the validation design by moving target thresholding, feature selection, and tuning inside the training folds.

Future work should use real platform data, stronger validation, refined predictive targets, and business guardrails before production use.

Source Snapshot

Main report	reports/final_visual_project_report.md
Cross-check report	reports/final_integrated_project_report.md
Repository	https://github.com/xur-2002/project

Sources: final visual report, integrated report, repository metric CSVs, and generated figures under **reports/outputs** and **reports/figures**. All metrics shown are read from repository outputs.

Created with assistance from ChatGPT